COURSE TITLE : BIOPHOTONICS
COURSE CODE : 6245
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction to Laser	15
2	Construction of LASERs	15
3	Biophotonic diagnostics	15
4	Biophotonic Therapy	15
TOTAL		60

Course General Outcome :

MODULE	G.O.	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To understand LASERS
	2	To understand LASER processes and features
2	3	To comprehend types of LASERs
3	4	To understand diagnostic applications of LASER
4	5	To understand therapeutic applications of LASER

G.O: General Outcome

On completion of the study the student will be able

MODULE I INTRODUCTION TO LASER

1.1.0 To understand LASERS

1.1.1 To define LASERs.
1.1.2 To explain stimulated emission, population inversion and gain
1.1.3 To explain three and four level lasers
1.1.4 To explain the properties of Laser
1.1.5 To define the terms Coherence, temporal and spatial coherence of laser radiation.

1.2.0 To understand LASER processes and features

1.2.1 To define Q-Switching
1.2.2 To list the Q-Switching devices
1.2.3 To define mode-locking.
1.2.4 To list the types of mode-locking
1.2.5 To explain Optical properties of the living tissues.
1.2.6 To define Photocoagulation, Photo thermal Ablation, Photochemical Ablation, Photo disruption, Photochemical Interaction
1.2.7 To explain thermal effects and mechanical effects of laser beam.

1.2.8 To explain Laser induced changes in tissue characteristics.
1.2.9 To explain Non thermal reactions of laser energy in tissue.
1.2.10 To explain holography and its application.

MODULE II CONSTRUCTION OF LASERS

2.1.0 To comprehend types of LASERs

2.1.1 To list different types of LASER.
2.1.2 To explain the Elements of LASER construction
2.1.3 To explain the construction and operation of Ruby laser.
2.1.4 To explain the construction and operation of Argon laser.
2.1.5 To explain the construction and operation of Helium-Neon laser.
2.1.6 To explain the construction and operation of CO2 laser.
2.1.7 To explain the construction and operation of Nd-YAG laser.

MODULE III BIOPHOTONIC DIAGNOSTICS

3.1.0 To understand diagnostic applications of LASER

3.1.1 To explain the principle and operation of flow cytometer
3.1.2 To list the applications of flow cytometry
3.1.3 To explain principle of optical biosensors
3.1.4 To explain bioimaging
3.1.5 To explain cellular, tissue and in vivo imaging.
3.1.6 To explain Optical Coherence tomography.

MODULE IV BIOPHOTONIC THERAPY

4.1.0 To understand therapeutic applications of LASER

4.1.1 To state the basic principle of Photodynamic therapy
4.1.2 To explain mechanism of photodynamic action
4.1.3 To list therapeutic applications of LASERs
4.1.4 To explain photosensitizers
4.1.5 To explain Laser tissue welding
4.1.6 To explain lasers in dermatology
4.1.7 To explain lasers in neurosurgery
4.1.8 To explain lasers in ophthalmology

CONTENTS

MODULE I LASERS

Definition of LASER-Stimulated emission- population inversion –Gain-Three and four level lasers-properties of Laser- Definition of Coherence, temporal and spatial coherence of laser radiation.

LASER processes and features

Definition of Q-Switching- Q-Switching devices-Definition of mode-locking-Types of mode-locking - Optical properties of the living tissues-Definition of Photocoagulation, Photo thermal Ablation, Photochemical Ablation, Photo disruption, Photochemical Interaction-Thermal effects and mechanical effects of laser beam - Laser induced changes in tissue characteristics- Non thermal reactions of laser energy in tissue-Holography and its application.

MODULE II TYPES OF LASER

Types of LASER-Elements of LASER construction- Construction and operation of Ruby laser, Argon laser, Helium-Neon laser, CO2 laser and Nd-YAG laser

MODULE III DIAGNOSTIC APPLICATIONS OF LASER

Principle and operation of flow cytometer-Applications of flow cytometry-Principle of optical biosensors - bio imaging - Cellular, tissue and in vivo imaging- Optical Coherence tomography

MODULE IV _THERAPEUTIC APPLICATIONS OF LASER

Principle of Photodynamic therapy- Mechanism of photodynamic action-Therapeutic applications of LASER-Photosensitizers- Laser tissue welding- Lasers in dermatology, neurosurgery and ophthalmology

REFERENCES

1. Handbook of Biomedical Instrumentation- RS Khandpur- Tata McGraw-Hill Pub., 2006
2. Lasers in Medicine- Leon Goldmann-Springer Verlag
3. The biomedical Laser Technology and clinical applications- Leon Goldmann-Springer Verlag
4. Introduction to Biomedical Photonics-Paras N Prasad-John Wiley 2003
5. Biomedical Photonics hand book-Tuan Vo Dinh-CRC Press-2003
6. Laser Fundamentals- William T Silfwast-Cambridge University press-1998